

AtlasMSI

Autoencoder-Based Representation Learning and Segmentation for MALDI-MSI

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2026-09-25

Welcome

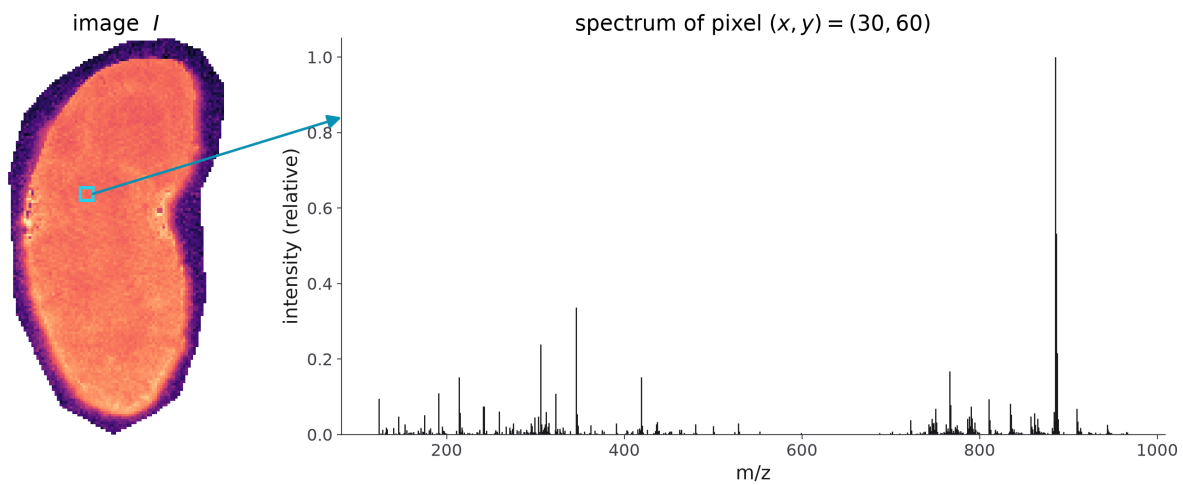
Welcome, BioGeMT attendees

Special thanks to Michał Ciach for the invitation.

Introduction

Data: mass spectrometry images (MALDI)

intuition



- every pixel of the image carries a whole mass spectrum
- MALDI: the tissue section is scanned pixel by pixel; one spectrum per position

Data: mathematical description

formal

Image $I \subset \mathbb{N}^2$ (or $\mathbb{N}^3$), pixel $p = (x, y)$

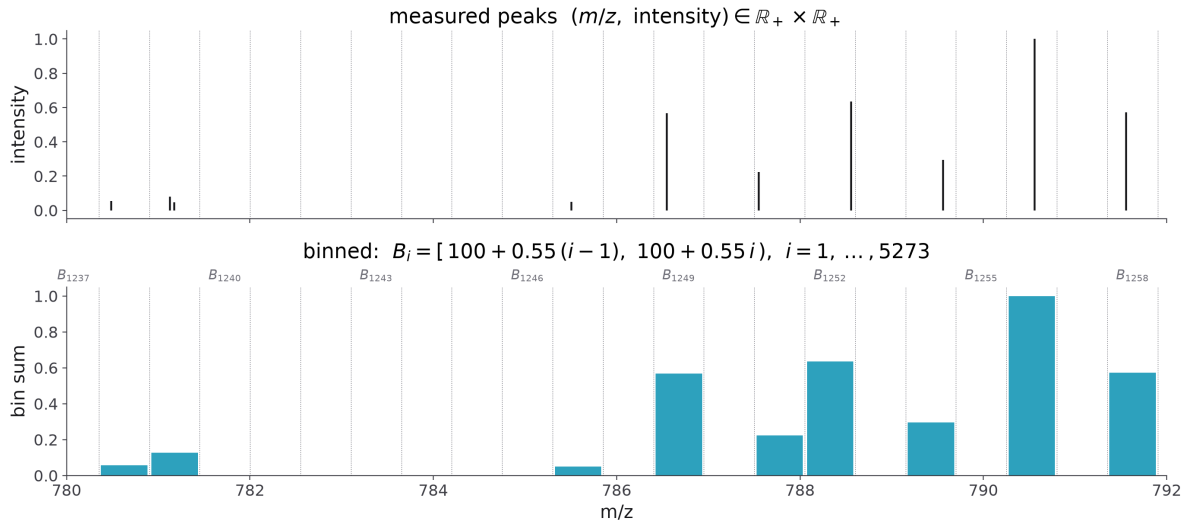
Spectrum at p — peaks $(m, v) \in \mathbb{R}_+ \times \mathbb{R}_+$

$$\mu_p = \sum_k v_k \delta_{m_k}$$

Binning of the m/z axis $[100, 3000]$ with step 0.55

$$B_i = [100 + 0.55(i-1), 100 + 0.55i), \quad i = 1, \dots, M$$

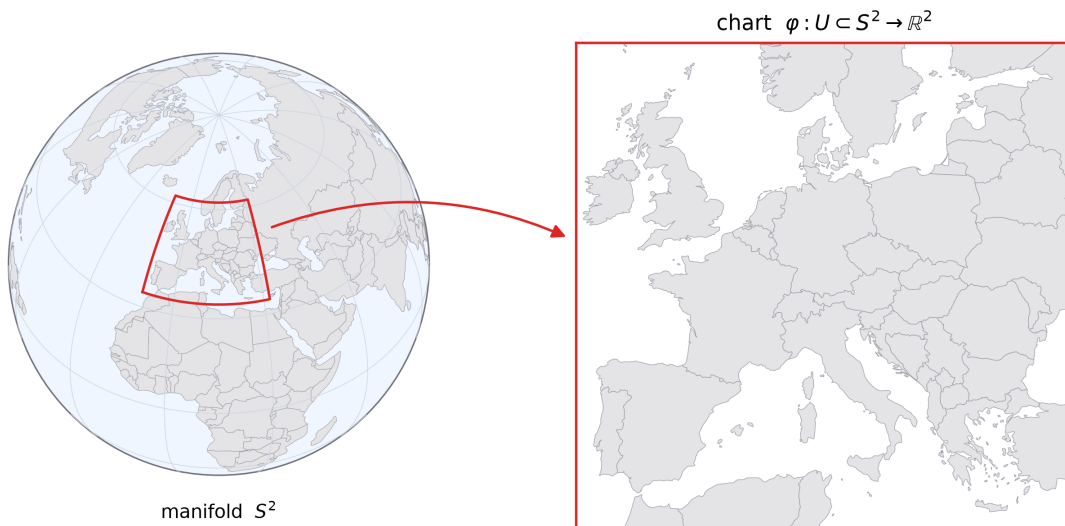
$$x_{p,i} = \sum_{m_k \in B_i} v_k, \quad M = 5273$$



$$\text{data} \subset I \times \prod_{i=1}^M (B_i \times \mathbb{R}_+) \cong I \times \mathbb{R}_+^M, \quad x_p \mapsto \frac{x_p}{\|x_p\|_1} \in \Delta^{M-1} \quad (\text{TIC})$$

Segmentation: where the word *atlas* comes from

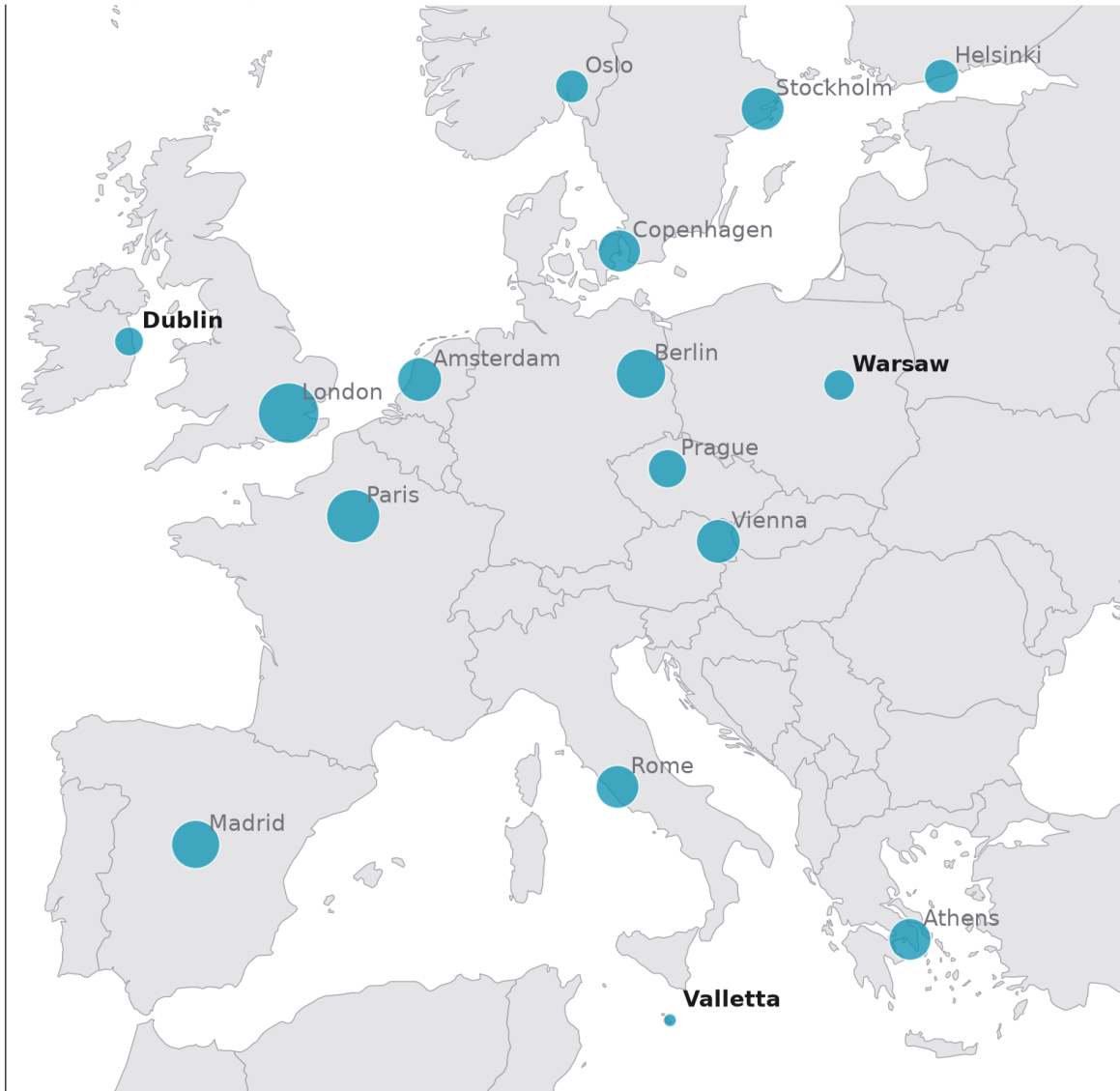
intuition



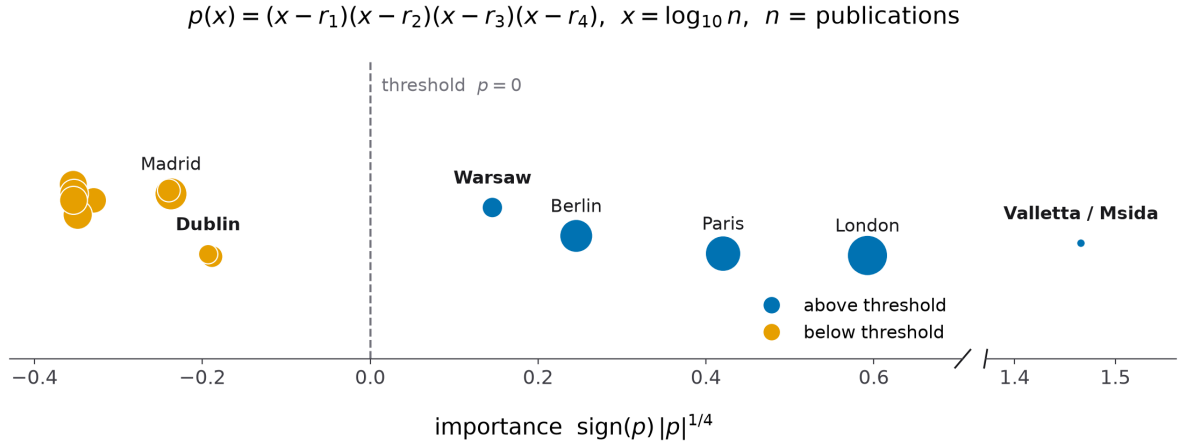
- the globe is known, but we work with **charts**: approximate, direct representations in $\mathbb{R}^2$
- an **atlas** is a collection of charts $(U_\alpha, \varphi_\alpha)$, $\varphi_\alpha: U_\alpha \rightarrow \mathbb{R}^d$

Segmentation: a trivial practical example

capitals: position $(\lambda, \phi) \in \mathbb{R}^2$, marker area = publications



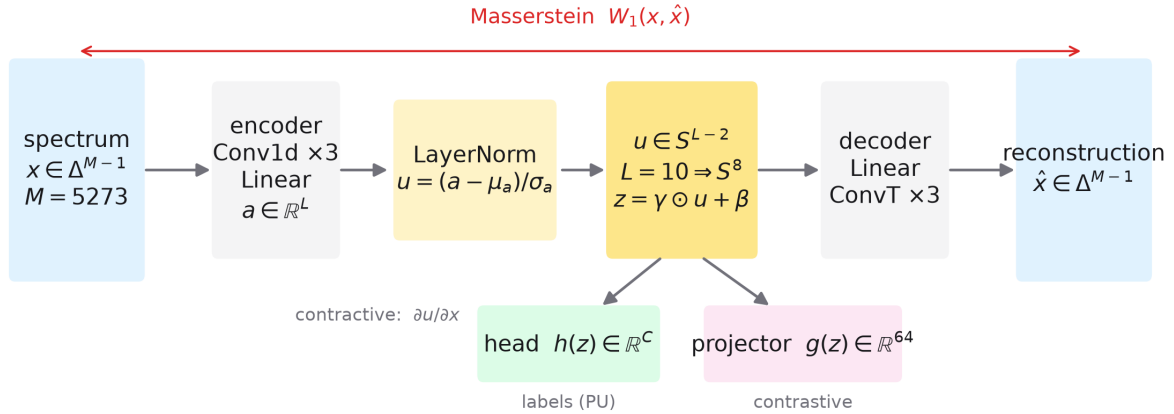
- a capital needs several coordinates (position, name, country, ...)
- we ask **one** question: how active is mass spectrometry there?
- PubMed publications with a capital-city affiliation, normalized over the shown capitals



- representation reduced to $\mathbb{R}$ — two groups appear

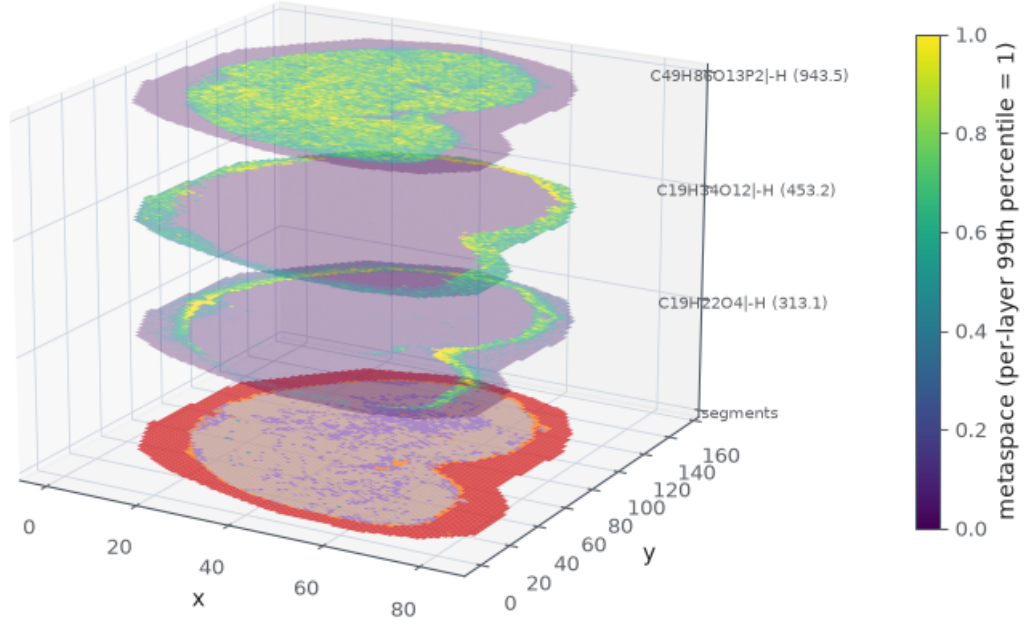
PubMed E-utilities, "mass spectrometry" [MeSH Terms] AND "<city>" [Affiliation], retrieved 2026-09-24.

Segmentation: the analysed example



- encoder compresses a spectrum into a latent point z
- decoder reconstructs the spectrum from z
- goal: construct z so that it is **stable** and **meaningful**

metaspace_spearman: 2024-02-20_01h46m58s, layers = metaspace



Segmentation: the atlas, formally

formal

$$\text{Enc} : \Delta^{M-1} \rightarrow S^{L-2}(\sqrt{L}) \subset \mathbf{1}^\perp \subset \mathbb{R}^L, \quad \text{Dec} : \mathbb{R}^L \rightarrow \Delta^{M-1}$$

Bottleneck LayerNorm of $a = \text{CNN}(x) \in \mathbb{R}^L$

$$u = \frac{a - \mu_a \mathbf{1}}{\sigma_a} \Rightarrow \mathbf{1}^\top u = 0, \quad \|u\|^2 = L$$

so u lies on the sphere $S^{L-2}(\sqrt{L})$ in $\mathbf{1}^\perp$; $z = \gamma \odot u + \beta$ is an affine image of it.

$$L = 10 \Rightarrow u \in S^8$$

Heads on the latent

- decoder $\hat{x} = \text{Dec}(z)$
- predictive head $h(z) \in \mathbb{R}^C$ (annotated ions)

- projector $g(z) \in \mathbb{R}^{64}$ (contrastive)

Data inside the sphere: neighbourhood-based (TwoNN) intrinsic dimension of the latent points ≈ 5.1 – 5.6

TwoNN on the canonical latent of the real-data baseline (axis 100–3000 m/z, 5 repetitions); notebook `part_1_06_latent_geometry`.

Components

The example we will develop

toy model

- two kidney sections of one series (METASPACE d28-2017-2, d28-2006-2): **16 230 pixels**
- m/z 200–950, step 0.55: **1364 bins; 21 annotated ions** (7 shared)
- same architecture and optimizer as the real model, **10 epochs**
- latent $L = 3 \Rightarrow u \in S^1$ (a circle), $L = 4 \Rightarrow u \in S^2$ (a sphere)
- components: **Masserstein, predictive head, contrastive, contractive**
- loss weights as in the real model: head 0.2, contrastive 0.001, contractive 0.002

The toy latent, interactively

909.6

826.6

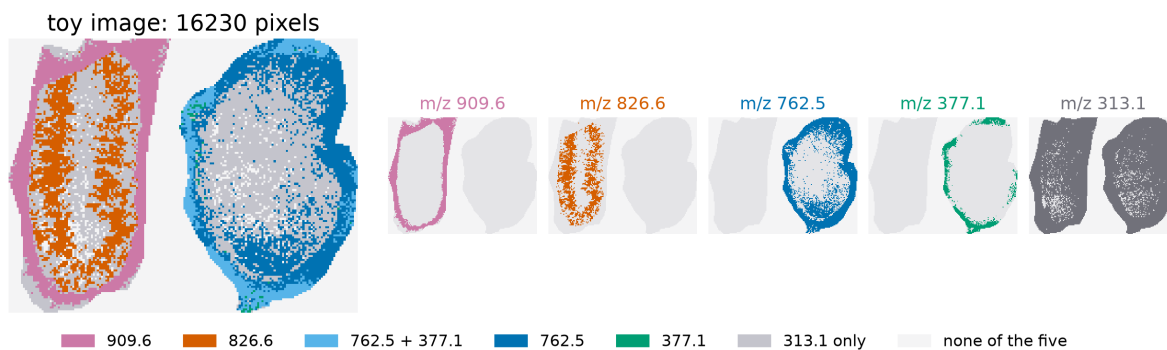
762.5 + 377.1

762.5

377.1

313.1 only

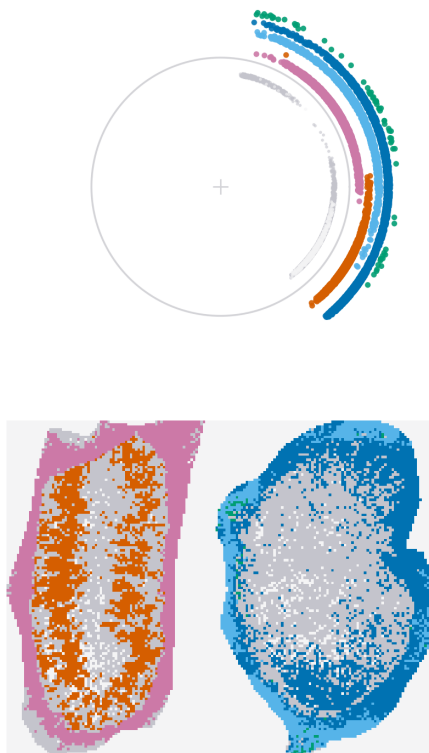
none of the five



Region key (left): 909.6 · 826.6 · 762.5 + 377.1 · 762.5 · 377.1 · 313.1 only · none of the five. **Ion maps (right):** panel title = m/z; ion-coloured pixels are annotated, grey pixels are not.

The toy latent: reconstruction only — S^1

reconstruction

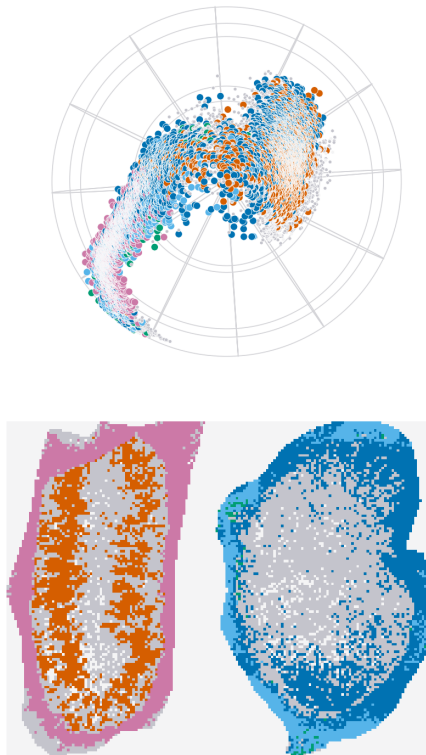


909.6
 826.6
 762.5 + 377.1
 762.5
 377.1
 313.1 only
 none of the five

top: every pixel on the circle S^1 (small radial jitter for visibility), colour = region; bottom: the image coloured by the position on the circle

The toy latent: reconstruction only — S^2

reconstruction



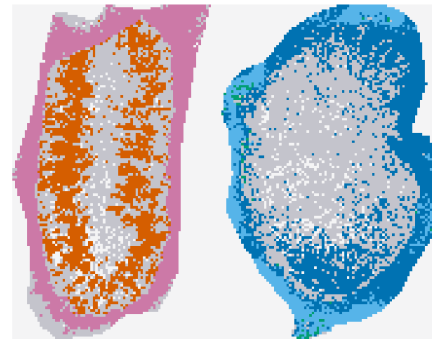
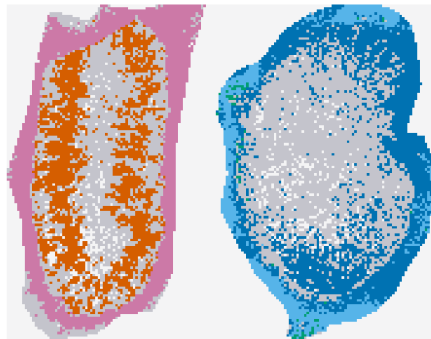
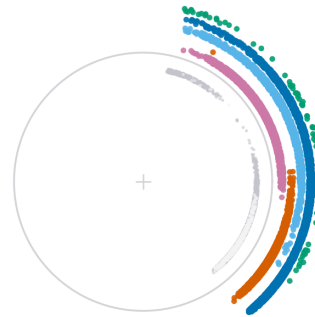
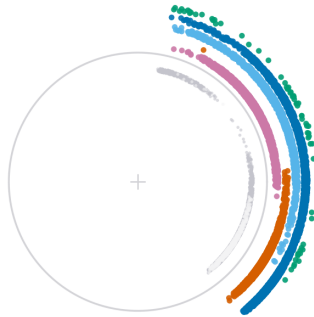
909.6 826.6 762.5 + 377.1 762.5 377.1 313.1 only none of the five

top: every pixel on S^2 , colour = region; bottom: the image coloured by the position on the sphere

Predictive head — S^1

reconstruction

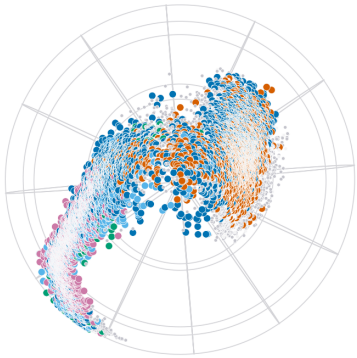
+ head



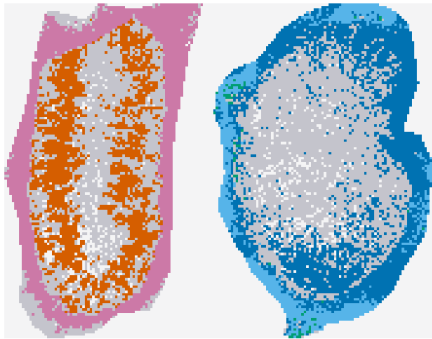
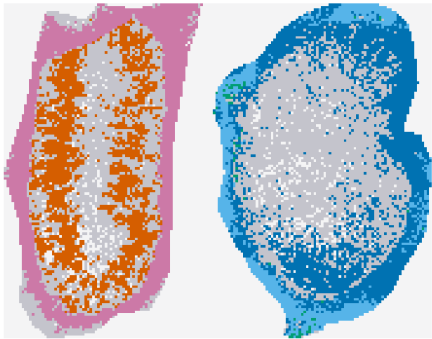
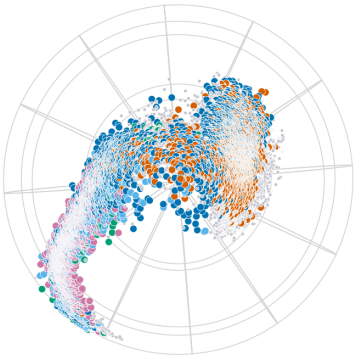
909.6
 826.6
 762.5 + 377.1
 762.5
 377.1
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 none of the five

Predictive head — S^2

reconstruction

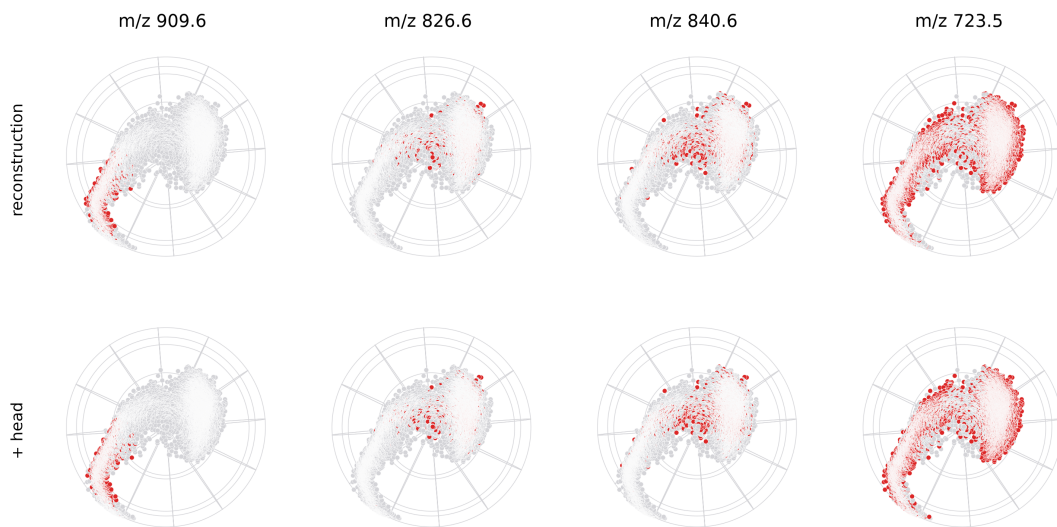


+ head



909.6 826.6 762.5 + 377.1 762.5 377.1 313.1 only none of the five

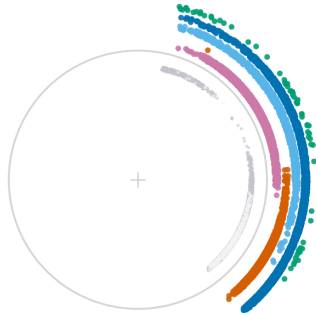
Predictive head: where the ions are



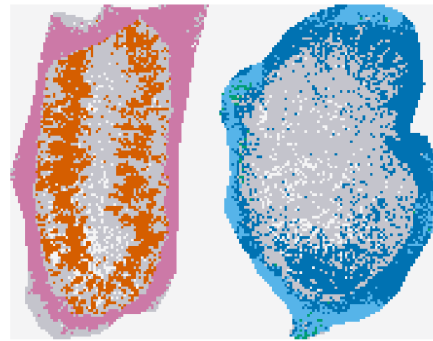
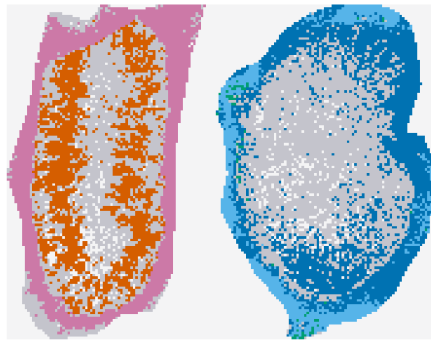
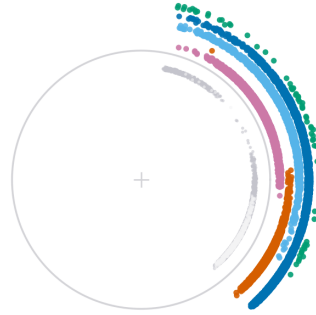
$\mathcal{L}_{\text{VPU}} = \frac{1}{C} \sum_c \left[\log \frac{1}{|P \cup U|} \sum_i \sigma(h_c(z_i)) - \frac{1}{|P_c|} \sum_{i \in P_c} \log \sigma(h_c(z_i)) \right]$ — positive-unlabelled: an unannotated ion is *unknown*, not absent

Contrastive learning — S^1

+ head



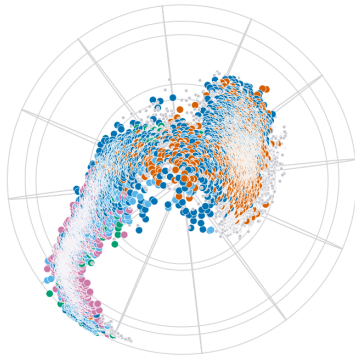
+ head + contrastive



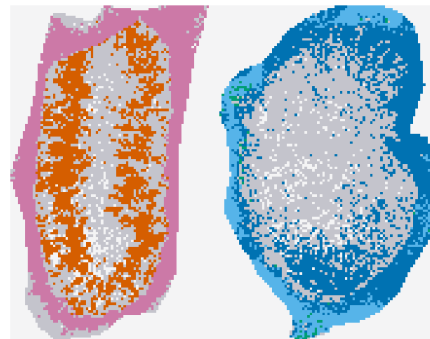
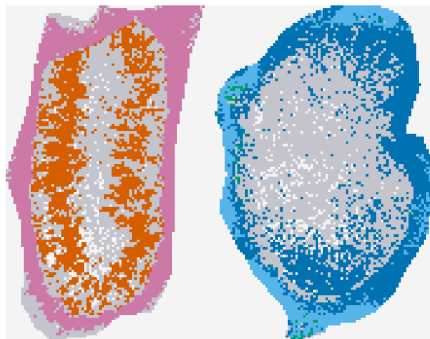
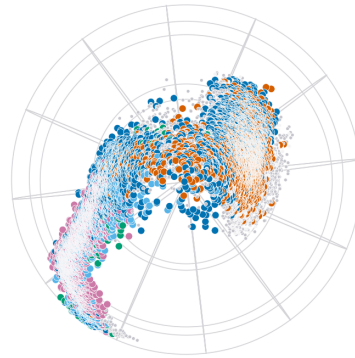
■ 909.6
 ■ 826.6
 ■ 762.5 + 377.1
 ■ 762.5
 ■ 377.1
 ■ 313.1 only
 ■ none of the five

Contrastive learning — S^2

+ head

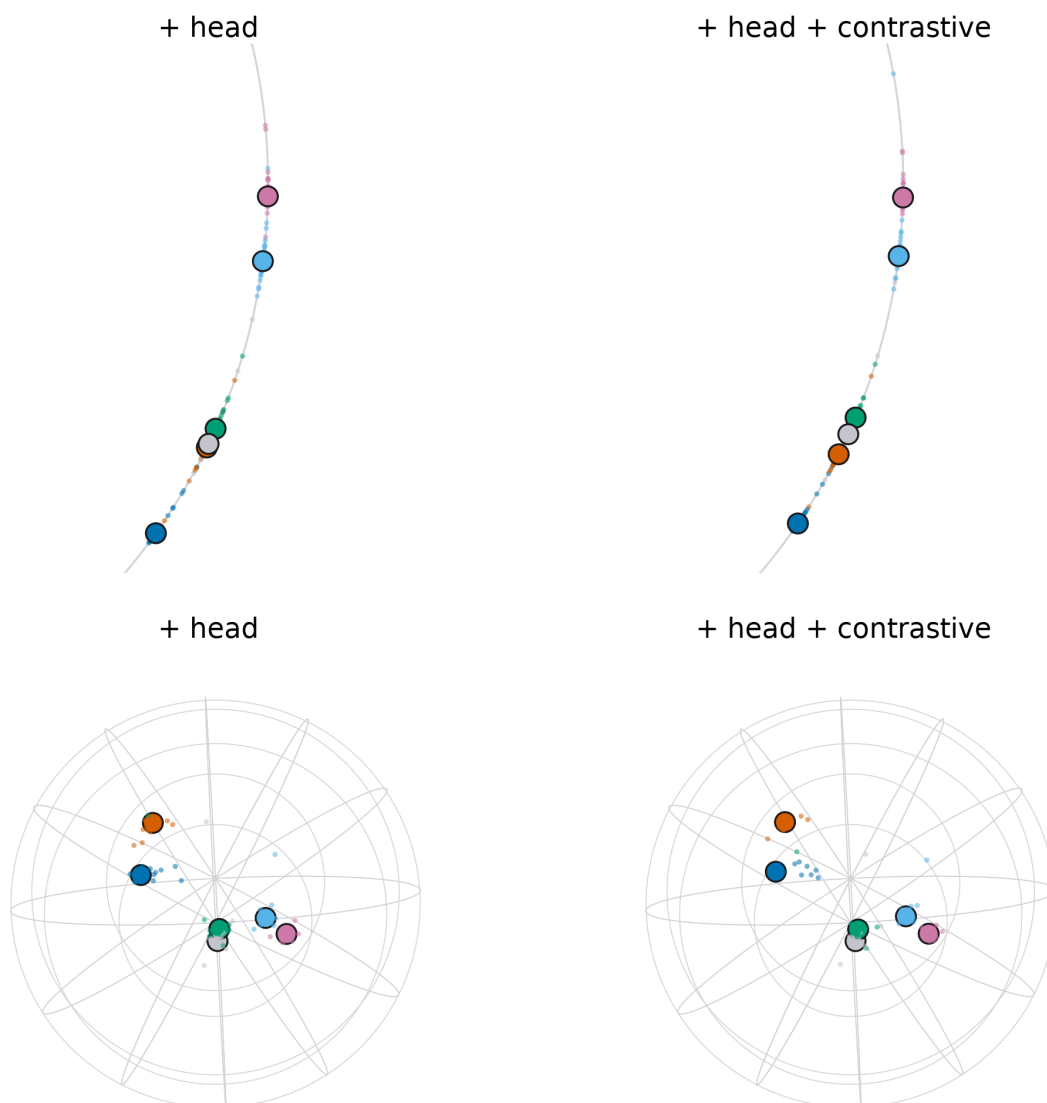


+ head + contrastive



■ 909.6
 ■ 826.6
 ■ 762.5 + 377.1
 ■ 762.5
 ■ 377.1
 ■ 313.1 only
 ■ none of the five

Contrastive learning: invariance to peak permutations



- a **view** $\tilde{x}$ permutes the shapes of three unannotated peak envelopes of x (TIC preserved)
- large points: pixels; small points: their views — the tighter the cloud, the more invariant the encoder

Contrastive learning: formally

formal

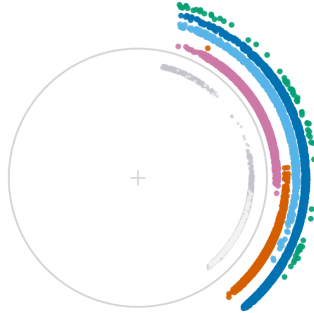
For a batch $x_1, \dots, x_B$ and views $\tilde{x}_i = T(x_i)$, projections $g_i = g(\text{Enc}(x_i))$, $g_{B+i} = g(\text{Enc}(\tilde{x}_i))$:

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{2B} \sum_{i=1}^{2B} \log \frac{\exp(\text{sim}(g_i, g_{\pi(i)})/\tau)}{\sum_{j \neq i} \exp(\text{sim}(g_i, g_j)/\tau)}, \quad \text{sim}(a, b) = \frac{a^\top b}{\|a\| \|b\|}, \quad \tau = 0.07$$

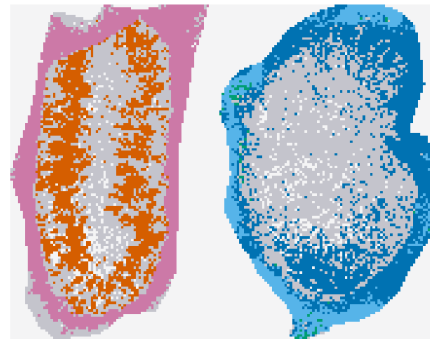
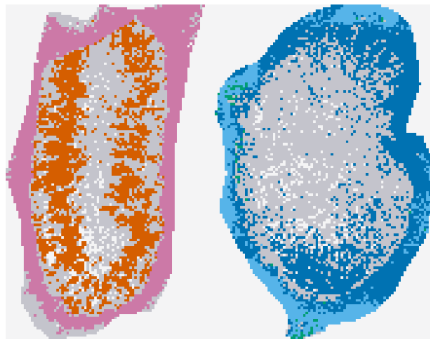
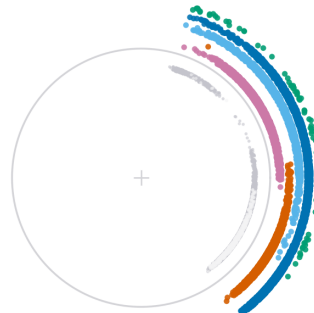
- $\pi(i)$: the other view of the same spectrum (positive pair); every other spectrum is a negative
- numerator **pulls** the two views together, denominator **pushes** different spectra apart
- T : cyclic permutation of peak-envelope shapes, **only peaks without annotation** — the representation may ignore unannotated rearrangements, but not the annotated evidence
- the loss acts on the projection $g(z)$, the latent z is shaped only through the gradient

Contractive learning — S^1

+ head



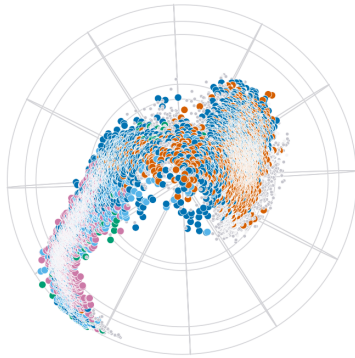
+ head + contractive



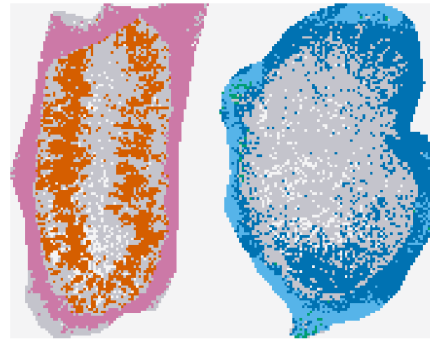
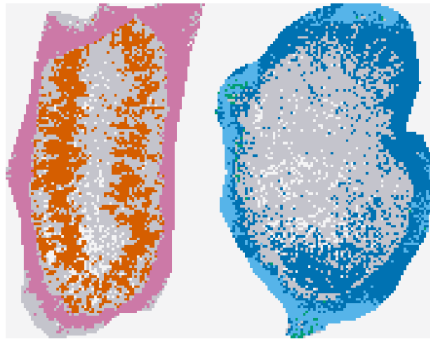
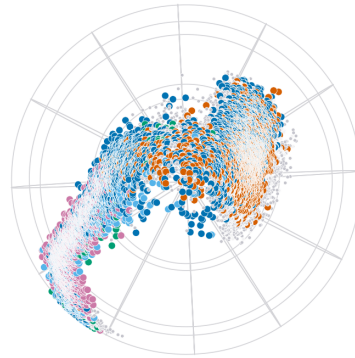
■ 909.6
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 ■ 762.5
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 ■ none of the five

Contractive learning — S^2

+ head

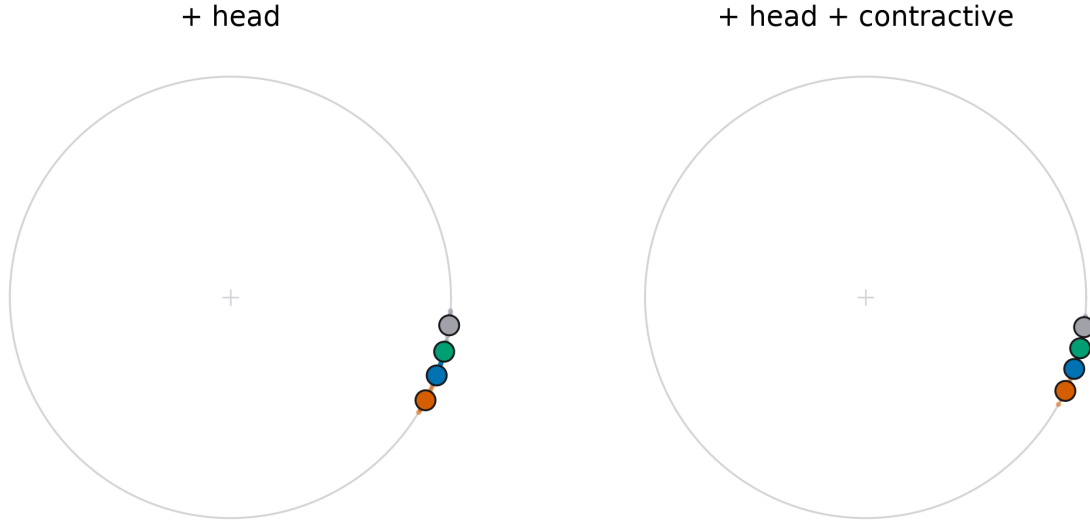


+ head + contractive

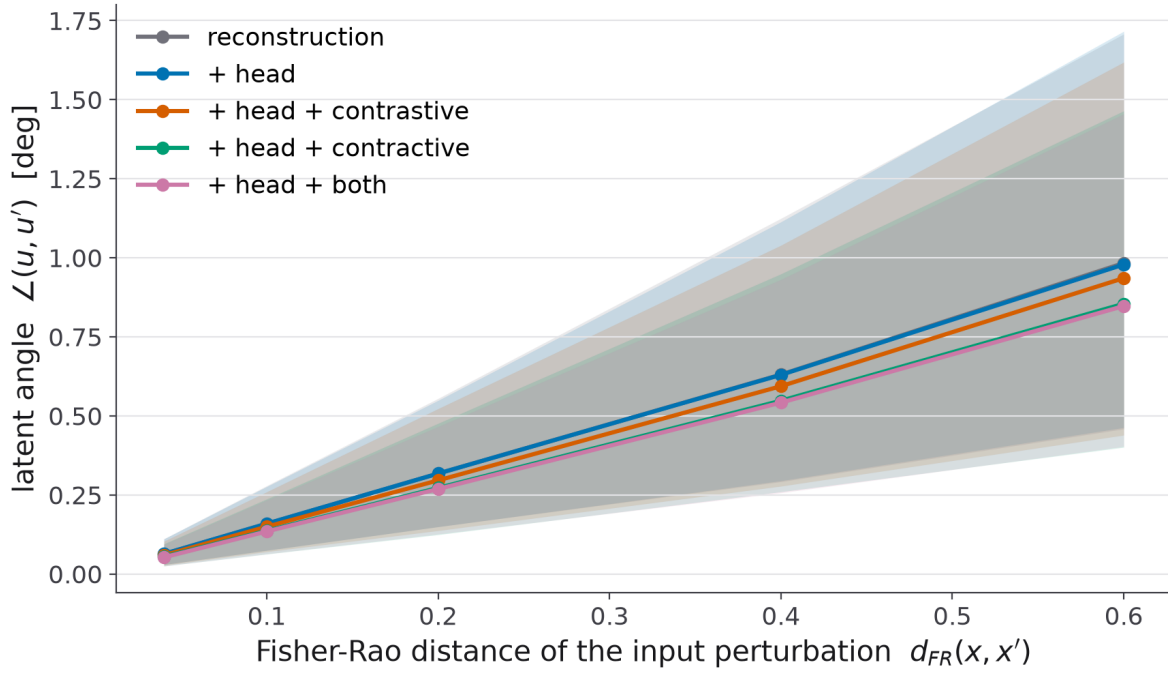


■ 909.6
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Contractive learning: sensitivity — S^1



large points = pixels, small points = copies perturbed by $d_{FR} = 0.6$

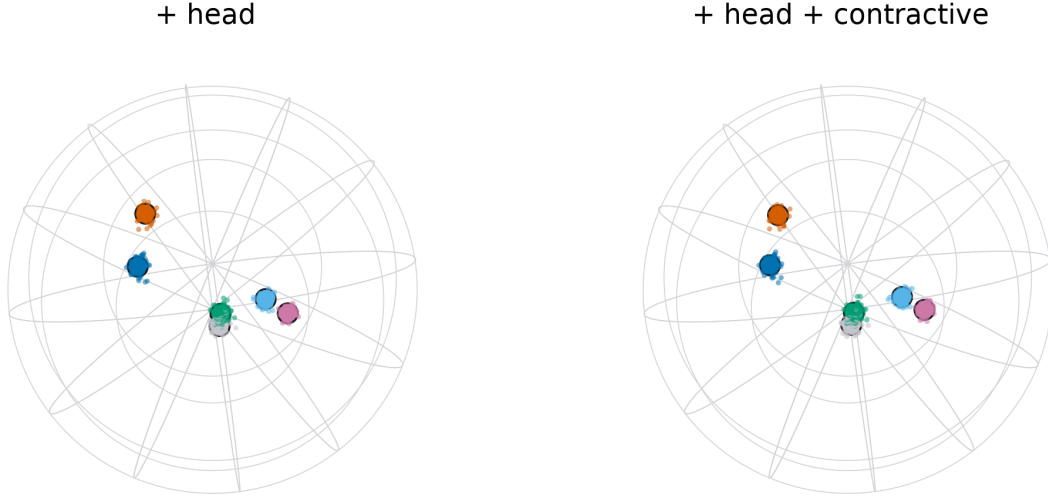


median rotation of the latent point (band: interquartile range) against the size of the input perturbation

- every input spectrum is moved by a fixed Fisher–Rao distance in random directions

- the rotation of its latent point measures the local sensitivity — lower = more stable
- S^1 toy trained on one section (854 pixels); the other toy figures use two sections.

Contractive learning: sensitivity — S^2



- large points: pixels; small points: copies perturbed by the same Fisher–Rao distance
- the tighter the cloud, the more stable the encoder

Contractive learning: formally

formal

Local sensitivity of the canonical latent: $J_u(x) = \partial u / \partial x \in \mathbb{R}^{L \times M}$

With $z = \gamma \odot u + \beta$:

$$J_z = \text{diag}(\gamma) \frac{1}{\sigma_a} P A_x, \quad P = I - \frac{1}{L} \mathbf{1}\mathbf{1}^\top - \frac{1}{L} u u^\top$$

$\|J_z\|$ can be reduced by shrinking γ without changing the encoder $\Rightarrow$ we penalize J_u .

Geometry of the input: spectra are points of the simplex; its natural metric is Fisher–Rao,

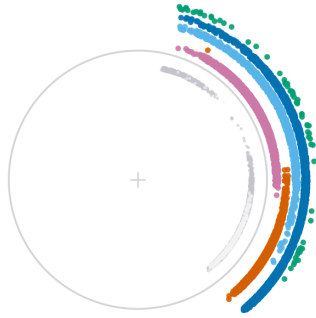
$$\langle v, w \rangle_x = \sum_i \frac{v_i w_i}{x_i}, \quad d_{FR}(x, x') = 2 \arccos \sum_i \sqrt{x_i x'_i}$$

$$A_x = J_u(x) \operatorname{diag}(\sqrt{x}) (I - \sqrt{x}\sqrt{x}^\top), \quad \mathcal{L}_{\text{contr}} = \sigma_{\max}(A_x)^2 + \alpha [\sigma_{\max}(A_x) - \tau]_+^2$$

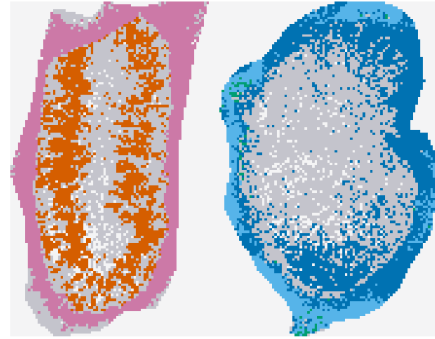
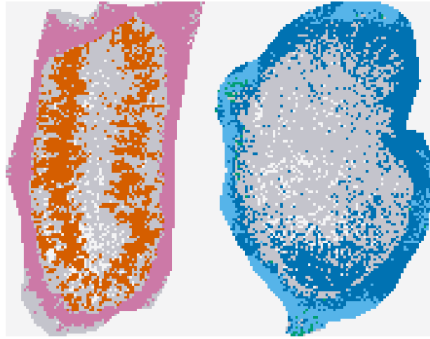
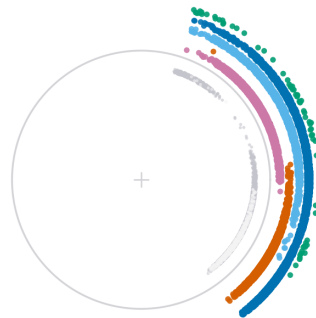
$\sigma_{\max}$ by power iteration; $\alpha = 2$, $\tau = 7$. Rifai et al. (2011) ICML — Frobenius, Euclidean, unnormalized latent.

Both regularizers — S^1

+ head



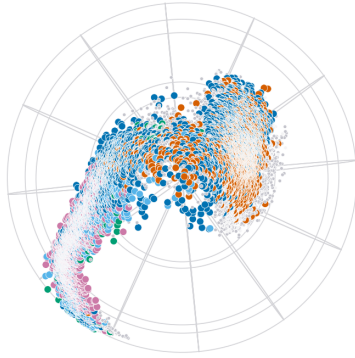
+ head + both



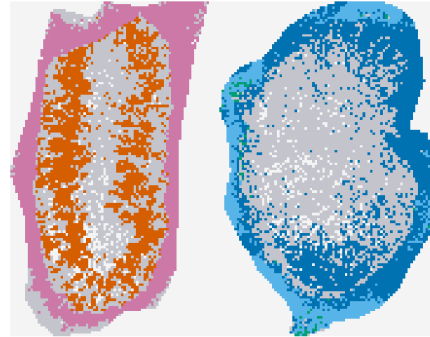
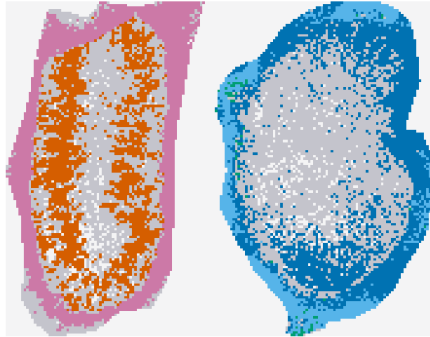
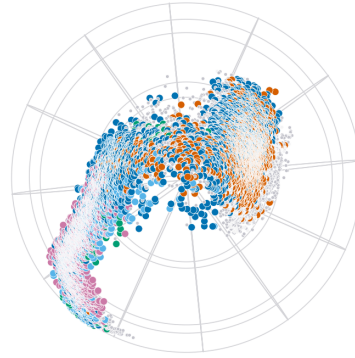
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Both regularizers — S^2

+ head

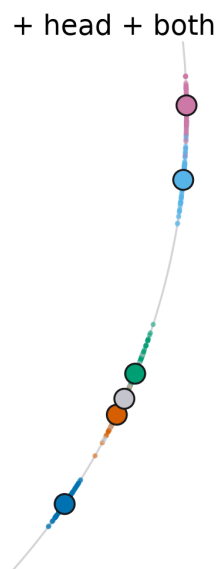
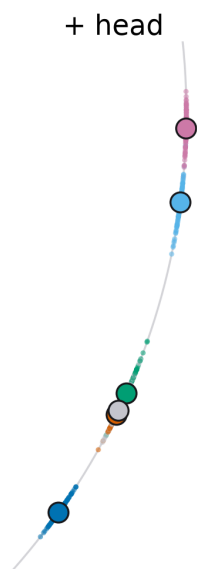


+ head + both

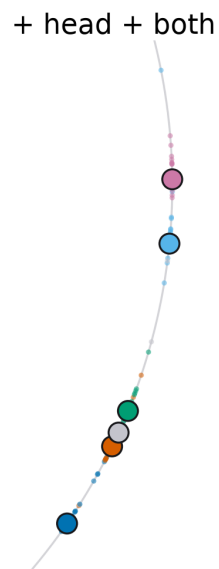


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Both regularizers: stability and invariance — S^1



input perturbed by $d_{FR} = 0.6$ (contractive view)

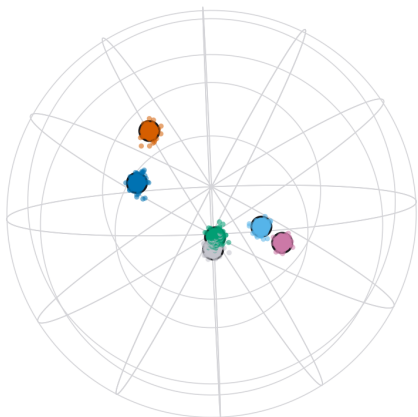


unannotated peaks permuted (contrastive view)

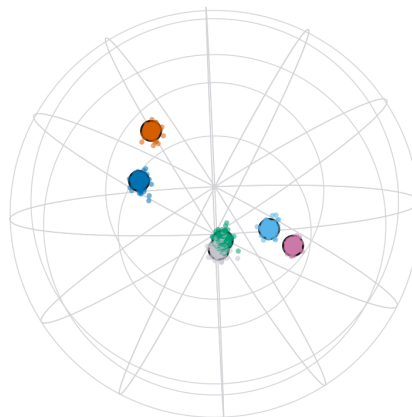
- large points: pixels; small points: perturbed copies (left) or permuted views (right)
- the tighter the cloud, the more stable (left) or invariant (right) the encoder

Both regularizers: stability and invariance — S^2

+ head

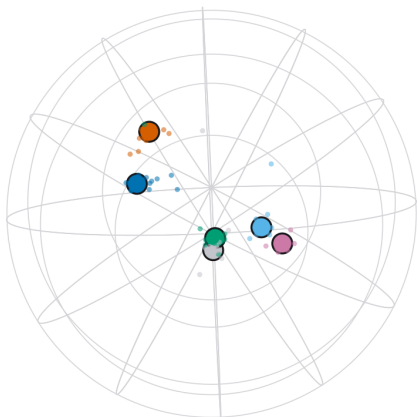


+ head + both

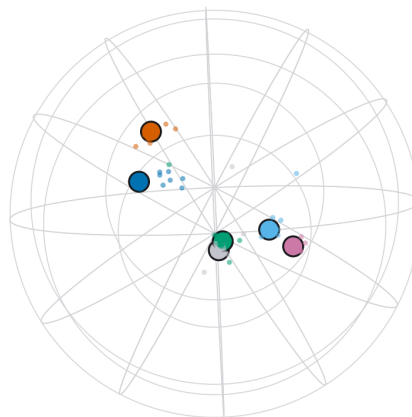


input perturbed by $d_{FR} = 0.6$ (contractive view)

+ head



+ head + both



unannotated peaks permuted (contrastive view)

- large points: pixels; small points: perturbed copies (left) or permuted views (right)
- the tighter the cloud, the more stable (left) or invariant (right) the encoder

Current state

Current state of the work

- representation model with predictive heads (variational PU) — prepared
- contrastive and contractive regularization — implemented and analysed
- synthetic pretraining — in progress
- next: dataset request and analysis on the benchmark dataset used by msiPL

msiPL: Abdelmoula et al. (2021) Nat. Commun. 12, 5544, doi:10.1038/s41467-021-25744-8

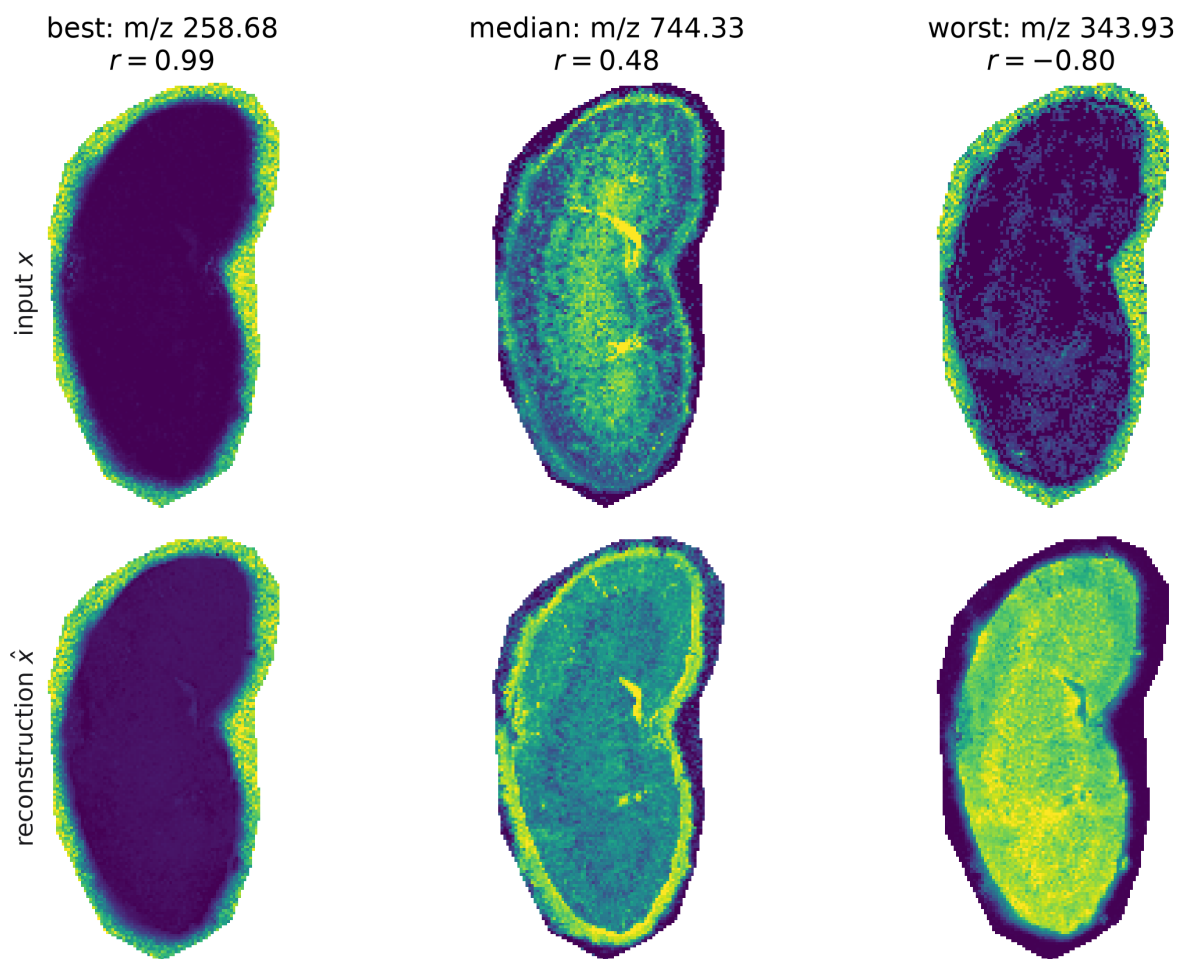
Masserstein: comparing spectra

TIC-normalized spectra are probability measures on the m/z axis. We compare them by the **Wasserstein distance** — the cost of moving intensity along m/z.

In one dimension it is exact and linear-time:

$$\begin{aligned} W_1(x, \hat{x}) &= \int |F_x(m) - F_{\hat{x}}(m)| dm \\ &= \sum_{i=1}^{M-1} \left| \sum_{j \leq i} (x_j - \hat{x}_j) \right| (m_{i+1} - m_i) \end{aligned}$$

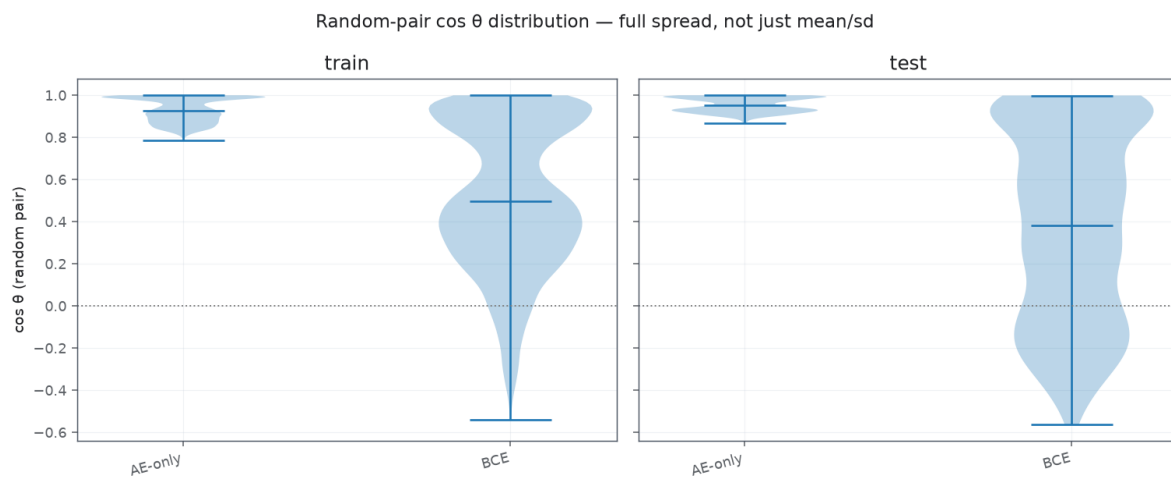
- no transport matrix, no iterative solver: $O(M)$
- a shifted peak costs its shift, not a full mismatch



real model, held-out image: best, median and worst reconstructed channel

Majewski, Ciach, Startek, Niemyska, Miasojedow, Gambin (2018) WABI, doi:10.4230/LIPIcs.WABI.2018.25
 · Ciach et al. (2020) Rapid Commun. Mass Spectrom., doi:10.1002/rcm.8956

How much of the latent sphere is occupied?



Random-pair $\cos \theta$ distributions for train and test. Values near 1 mean two latent directions are close; a broader spread toward 0 or below indicates wider angular coverage of the sphere.

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